

THE CHALLENGES IN COMPUTING

Digital Divide

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It is the gap between those with Internet access and those without it. It is one of the biggest challenges that areas of computing share as one would need Internet access to utilize computing, computers, networks, and the web.

The disparities in Internet access are attributed to the following factors:

1. **Availability** – It deals with the accessibility of the Internet in the area without going the extra mile, such as climbing on the roof or a mountain for connectivity.
2. **Affordability** – It concerns whether the Internet is affordable or not based on a household's income.
3. **Quality of Service** – deals with the quality of upload and download speed given the price and based on user needs.
4. **Relevance** – concerns the need for Internet in local areas and if there is local interest and understanding of the relevance of Internet access.

Data on Digital Divide

There are several approaches to measure the gaps in Internet access. Some sources look at the saturation of devices used, while others look at the areas covered by Internet Service Providers (ISPs).

According to the UN's Broadband Commission for Sustainable Development, an estimated 3.6 billion people remain entirely unconnected to the Internet, which is about 53.6% of the global population as of 2022.

Specific Impacts of Digital Divide

Most people with lower incomes face the adverse impacts of the digital divide. The lack of Internet access creates challenges that hinder economic opportunities and adaptability.

Here are some of the adverse impacts of the digital divide:

1. **Healthcare opportunities:** Health experts argue that Internet access is a significant public health issue stating that it is a social factor of health as lack of access means exclusion from vital health information and resources.
2. **Economic opportunities:** The lack of strong and quality broadband connections decreases economic chances and adaptability for workers and business owners in e-commerce.
3. **Educational opportunities:** The lack of Internet access for students denies them the educational benefits of the Internet. This gap results in unequal learning outcomes and has been particularly proven during the COVID-19 pandemic.

Solutions to Bridging the Digital Divide

Governments and private organizations have successfully brought Internet access to as many people as possible, but there is no single solution to bridging the gap.

The other unconnected half of the world's population is the hardest to connect to the Internet as they often live in low-density or low-income areas that are not feasible for telecommunication companies to provide services and build infrastructures.

Community networks solve these challenges by bringing affordable Internet access to those most in need. But beyond making the Internet accessible and affordable, community networks help address some of the specific impacts of the digital divide.

For example, the Internet Society in Nepal initiated *Effective Broadband for Health* which aims to help health problems by bringing Internet access to their communities by ensuring affordable telehealth options in rural areas.

Murambinda community network in Zimbabwe has brought changes in education, healthcare, and agriculture through Internet access. It shows that community members who build networks also strengthen their shared digital literacy skills.

Digital Inequality

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It is the discrepancy in knowledge and skill in using computing and Internet technologies among individuals with socioeconomic backgrounds, IT experiences, and varied demographics. It is one of the most critical issues for a country's social and economic growth.

Digital inequality resists the continuation of national development and prevents a nation from becoming a developing country.

Digital Inequality vs. Digital Divide

Digital Divide contrasts the “haves” and “have-nots” of proper Internet connectivity. At the same time, Digital Inequality highlights the differences between the users based on their IT skills that reduce the digitization process, its deployment, and adoption of any developing country.

For example, students unable to participate in online classes because of limited Internet access or unavailable smartphone or computer create a digital divide between those students who have it and those who do not. On the other hand, it results in digital inequality when a student has Internet access and devices but does not know how to utilize them for reasons such as face-to-face preferability and limited technological skills.

Elements of Digital Inequality

These are the elements that enforce the existence of digital inequality in a community or a nation.

1. **Educational** – focuses on educating the citizens to enhance their computing and Internet technologies skills to use them properly. Factors such as lack of basic education, non-proficiency in technology, and lack of skills in accessing the Internet and devices contribute to this problem.
2. **Infrastructural** – facilitates issues that help citizens to have easy and effective access to information technologies. Consistent and reliable Internet towers and power supply in rural and remote parts must be provided to establish accessibility to computing and Internet technologies.
3. **Social** – focuses on the citizens' trust and awareness about social issues and uses computing and Internet technology. Rural and illiterate citizens are unaware nor motivated and not getting enough support to learn and use digital services for their economic development
4. **Economical** – highlights the importance of computing and Internet technology from an economic standpoint. It includes the lack of digitization-based investment and limited opportunity for online-based business. Rural and illiterate citizens are unwilling to invest in digitization-based businesses, resulting in e-commerce sites' demise.
5. **Usable Design** – deals with how easily citizens use computing and Internet technologies. User experience (UX), user interface (UI) in multiple languages, and accessibility are appropriately designed. Rural and illiterate citizens are unwilling to use new technologies with difficult usability and user experience.

Solutions for Digital Inequality

It is important to adopt, accept, and use computing and Internet technologies to achieve the nation's goal of being an established country centered on digitalization. Challenges that come with digital inequality and the digital divide must be addressed and overcome to make that possible.

Digital Equity and **Digital Inclusion** are vital to resolve the challenges.

Digital Equity is the condition wherein individuals and communities have the information technology capacity for full participation in society, democracy, and the economy. It is significant for civic and cultural duties, employment, and access to essential services.

Digital Inclusion is necessary to ensure that individuals and communities, including the least privileged, have access to information and communication technologies. Digital inclusion strategies must evolve as rapidly as the development and changes in technology.

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